

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

# Chapter 21: Trigonometrical Identities

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 The three basic identities

1. Assertion (A): If  $\sin^2 A + \sin A = 1$ , then  $\cos^4 A + \cos^2 A = 1$ . Reason (R):  $1 - \sin^2 A = \cos^2 A$ . (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**

**SOLUTION**

From the given relation,  $\sin A = 1 - \sin^2 A = \cos^2 A$  (using R).

Then  $\cos^4 A + \cos^2 A = \sin^2 A + \sin A = 1$ . So A is true, and R is exactly what proves it.

**Answer: (c)**

## 2 Proving identities

2. Prove that  $\frac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$ . **ICSE 2025, 4 marks**

**SOLUTION**

$$\cot A + \tan A - 1 = \frac{\cos^2 A + \sin^2 A - \sin A \cos A}{\sin A \cos A} = \frac{1 - \sin A \cos A}{\sin A \cos A}$$

$$\sin^3 A + \cos^3 A = (\sin A + \cos A)(\sin^2 A - \sin A \cos A + \cos^2 A) = (\sin A + \cos A)(1 - \sin A \cos A)$$

$$\text{LHS} = \frac{(1 - \sin A \cos A)(\sin A + \cos A)}{\sin A \cos A (\sin A + \cos A)(1 - \sin A \cos A)} = \frac{1}{\sin A \cos A}$$

$$= \sec A \operatorname{cosec} A = \text{RHS. Proved.}$$

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